A APPENDIX

B ALGORITHM PSEUDOCODE

Algorithm 1 details the per-frame execution of the RSGM Data Plane (§5). Lines 1–6 implement Operator 1 (Geometry-Guided Image Reduction, §5.1): the point cloud is projected onto the image plane (Step 1), expanded into a depth-stratified relevance mask (Step 2), and used to suppress high-frequency content in non-target regions before codec encoding (Step 3). Lines 7–15 implement Operator 2 (Semantic-Guided Point Cloud Sampling, §5.2): a Laplacian-based saliency map is computed from the image (Step 1), a content-adaptive quadtree is constructed (Step 2), and points are retained per leaf node at a saliency-modulated rate controlled by $\theta_P^{(t)}$ (Step 3).

Algorithm 1 RSGM: Reciprocal Semantic-Geometric Modulation

Require: Image $\mathbf{I}_t$, Point Cloud $\mathbf{P}_t$, Control knobs $\boldsymbol{\theta}^{(t)} = \{\theta_I^{(t)}, \theta_P^{(t)}\}$
Ensure: Compressed pair $(\tilde{\mathbf{I}}_t, \widehat{\mathbf{P}}_t)$

— Operator 1: Geometry-Guided Image Reduction —

1: $\mathcal{U}_t \leftarrow \textsc{Project}(\mathbf{P}_t, \mathbf{K}, \mathbf{R}, \mathbf{t})$ ▹ 3D-to-2D alignment
2: Partition $\mathcal{U}_t$ into depth strata $\{\mathcal{S}_0, \ldots, \mathcal{S}_{K-1}\}$
3: $\mathbf{M}_t \leftarrow \bigcup_k (\mathcal{S}_k \oplus \mathbf{B}_k)$ ▹ Depth-stratified expansion
4: $\mathbf{I}_{LF} \leftarrow \mathbf{I}_t * G_\sigma$
5: $\tilde{\mathbf{I}}_t \leftarrow \mathbf{M}_t \odot \mathbf{I}_t + (1 - \mathbf{M}_t) \odot \mathbf{I}_{LF}$ ▹ Spectral filtering
6: Encode $\tilde{\mathbf{I}}_t$ with quality parameter $\theta_I^{(t)}$

— Operator 2: Semantic-Guided Point Cloud Sampling —

7: $\mathbf{S}_t \leftarrow |\nabla^2 \mathbf{I}_t| \;/\; (\max |\nabla^2 \mathbf{I}_t| + \epsilon)$ ▹ Laplacian saliency
8: $\mathcal{T} \leftarrow \textsc{BuildQuadtree}(\mathbf{P}_t, \mathbf{S}_t, \sigma_{th}, N_{min}, D_{max})$
9: $\bar{S} \leftarrow W_{total} \;/\; |\mathbf{P}_t|$ ▹ Global mean saliency
10: $\widehat{\mathbf{P}}_t \leftarrow \emptyset$
11: **for** each leaf $q \in \text{Leaf}(\mathcal{T})$ **do**
12: $\quad \bar{S}_q \leftarrow W_q / |P_q|$; $\rho_q \leftarrow \min(1,\ \theta_P^{(t)} \cdot \bar{S}_q / \bar{S})$
13: $\quad N_q \leftarrow \lfloor |P_q| \cdot \rho_q \rfloor$
14: $\quad \widehat{\mathbf{P}}_t \leftarrow \widehat{\mathbf{P}}_t \;\cup\; \textsc{Sample}(q.\text{bucket},\ N_q)$
15: **end for**
16: **return** $(\tilde{\mathbf{I}}_t,\ \widehat{\mathbf{P}}_t)$

Algorithm 2 describes the three-stage CRAO Control Plane (§6). Stage 1 (Line 1) instantiates the data-condition state vector $\boldsymbol{\Phi}_t$ using the lightweight proxies from §6.1. Stage 2 (Lines 2–3) generates a temporally regularized candidate set anchored at the previous plan $\boldsymbol{\theta}^{(t-1)}$ (Definition 1), supplemented by uniform exploration samples for robustness. Stage 3 (Lines 4–14) evaluates each candidate using the hybrid cost-utility models from §6.2—the latency model f_{lat} filters infeasible plans, and the utility model f_{acc} selects the best feasible one. If no candidate meets the deadline, the fallback (Lines 15–17) selects the minimum-latency configuration to ensure graceful degradation.

Algorithm 2 CRAO: Collaborative Rate-Allocation Optimization

Require: Raw frames $(\mathbf{I}_t, \mathbf{P}_t)$, Previous plan $\boldsymbol{\theta}^{(t-1)}$, Bandwidth B_t, Latency threshold τ_{target}

Ensure: Optimal plan $\boldsymbol{\theta}^{(t)} = \{\theta_I^{(t)}, \theta_P^{(t)}\}$

— **Stage 1: State Sketching** —

$\boldsymbol{\Phi}_t \leftarrow [\phi_{\text{tex}}, \phi_{\text{den}}, \phi_{\text{dyn}}^{\text{img}}, \phi_{\text{dyn}}^{\text{pc}}]$ from $(\mathbf{I}_t, \mathbf{P}_t)$

— **Stage 2: Candidate Generation (Def. ??)** —

$\boldsymbol{\Theta}_t \leftarrow$ Sample $(1-\epsilon) \cdot n$ candidates via $P \propto e^{-\beta \|\boldsymbol{\theta} - \boldsymbol{\theta}^{(t-1)}\|_1}$

$\boldsymbol{\Theta}_t \leftarrow \boldsymbol{\Theta}_t \cup \textsc{UniformSample}(\boldsymbol{\Theta}, \epsilon \cdot n)$

— **Stage 3: Constrained Plan Selection** —

$u_{\max} \leftarrow -\infty;\ \boldsymbol{\theta}^{(t)} \leftarrow \boldsymbol{\theta}^{(t-1)};$ `feasible` $\leftarrow$ `false`

for each $\boldsymbol{\theta} \in \boldsymbol{\Theta}_t$ **do**

  $\widehat{\mathcal{T}} \leftarrow \max_m \left(f_{\text{comp}}^{(m)}(\boldsymbol{\Phi}_t, \theta_m) + f_{\text{trans}}^{(m)}(\boldsymbol{\Phi}_t, \theta_m, B_t)\right)$

  if $\widehat{\mathcal{T}} \leq \tau_{\text{target}}$ **then**

    `feasible` $\leftarrow$ `true`

    $\widehat{\mathcal{U}} \leftarrow f_{\text{acc}}(\boldsymbol{\Phi}_t, \boldsymbol{\theta})$

    if $\widehat{\mathcal{U}} > u_{\max}$ **then**

      $u_{\max} \leftarrow \widehat{\mathcal{U}};\ \boldsymbol{\theta}^{(t)} \leftarrow \boldsymbol{\theta}$

    end if

  end if

end for

— **Fallback** —

if not `feasible` **then**

  $\boldsymbol{\theta}^{(t)} \leftarrow \arg\min_{\boldsymbol{\theta} \in \boldsymbol{\Theta}_t} \widehat{\mathcal{T}}_{total}(\boldsymbol{\theta})$

end if

return $\boldsymbol{\theta}^{(t)}$